Fiscal Year:	FY 2024	**Task Last Updated:**	FY 08/12/2024
PI Name:	Khodadad, Christina Ph.D.		
Project Title:	Microbe-Plant Interactions with International Space Station Microbial Communities in Veggie Flight Crops		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Microbiology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	32931	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2018 Space Biology (ROSBio) NNH18ZTT001N-FG2. App D: Flight and Ground Space Biology Research
Start Date:	10/01/2021	**End Date:**	09/30/2023
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA KSC
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Flight Program:			
Flight Assignment:	End date changed to 09/30/2023 per NSSC information (Ed., 7/3/23)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Singh, Nitin Ph.D. (Jet Propulsion Laboratory) Simpson, Anna C. (JPL)		
Grant/Contract No.:	80NSSC21K0668		
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Performance Goal Text:			

Task Description:	The ability to grow safe, fresh food as a dietary supplement for astronauts in space is an important goal for NASA. Food crops such as leafy greens grown in space experience different environmental conditions than plants grown on Earth (i.e., reduced gravity, elevated radiation levels, CO2 levels, etc.). These conditions also may have a profound effect on the microbial communities that are an integral part of the plants' microbiomes. Plants and their microbiomes may also have an effect on their surrounding environment. The NASA Vegetable Production System (Veggie) is open to the International Space Station (ISS) atmosphere so interactions and exchanges of microbes from the ISS and the crew to the plants and the converse is probable. An in-depth genetic analysis of archived samples and bacterial and fungal isolates would provide the information necessary to correlate the Veggie microbial communities to other ISS sources (air, surfaces, crew members, and water) that are currently being studied in part by the Microbial Tracking investigations at Jet Propulsion Laboratory (JPL) and microbial monitoring activities at NASA Johnson Space Center (JSC). The goals of this study is to expand our knowledge of how microbiological organisms and/or communities associated with Veggie plants acclimate to and change over time in the spaceflight environment. This study will also contribute to the evaluation of the benefits for potential manipulation and control in the ISS or other closed environments. An additional goal would be to characterize spaceflight's effects on microbial community dynamics by investigating the mixed community of representative microbes found on station and in Veggie to study how they may compete with one another, and thrive in air, water, on surfaces, and in media. This study provides a means to investigate the biology of established communities that are found on spacecraft and their interactions as defined in Sub-topic MB-A of the solicitation. [Ed. Note: Per the 2018 Space Biology (ROSBio) NNH18ZTT001N-FG2 solicitation: Studies that characterize spaceflight's effects on microbial community dynamics by mixing combinations of representative microbes to study how they interact, synergize, compete with one another, and thrive in air, water, on surfaces and in soil. Studies may investigate the biology of established communities that are found on spacecraft.] These tasks will be completed using existing nucleotides, DNA, isolated from Veggie crops previously grown on the ISS and ground controls to include, but not limited to: VEG01 (A-C) through VEG03 (A-F). DNA will be labeled for metagenomics and/or whole genome sequencing. Shotgun metagenomics, the untargeted sequencing of all microbial genomes present in a sample, provides information on the structure and function of a microbial community. Shotgun sequencing can be used to profile the taxonomic composition and functional potential of microbial communities and to recover whole genome sequences. The benefits provided by substrate and seed microbial communities are important for the sustainability of the substrate and for the health of the plants themselves. Microbes in both the rhizosphere and in the phyllosphere have been found to contribute to nutrient cycling and nutrient flow as well as carbon sequestration, protection and defense from plant pathogens, and general plant health and growth. Identifying and understanding the microbiome of the crop seeds and plants during developmental stages will provide necessary information to maintain a healthy, nutritious vegetable source. Data from this study may also show the relationship between the environmental microbial community and the Veggie plant communities.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Shotgun sequencing can be used to profile the taxonomic composition and functional potential of thousands of microorganisms in parallel and sample genes of the existing microbial communities. This provides insight into community diversity and functional capacity. Understanding the functions and characterizing specific strains of these communities can determine the contributions of microorganisms to plant, human, as well as environmental health. The benefits provided by substrate media, microbial communities are important for the sustainability of the substrate and for the health of the plants themselves. Microbes in both the rhizosphere (root zone) and in the phyllosphere (plant) have been found to contribute to nutrient cycling and nutrient flow as well as carbon sequestration, protection and defense from plant pathogens and general plant health and growth. Identifying and understanding the microbiome of the plants and their environments will provide necessary information to maintain a healthy, nutritious vegetable source to supplement crew diets for astronauts aboard the International Space Station (ISS) for future long-duration missions. It is also important that the effects of environmental factors be investigated and determined to be able to maintain a healthy crop of fresh fruit and vegetables on the station and for future long-duration missions. Characterization of the microbial community provides a fundamental baseline knowledge of how the phyllosphere (plant) and rhizosphere (root) microbiomes adapt under spaceflight conditions and what benefit these microbes may play in plant health and growth. Data obtained may also provide information on plant grow promoting microbes that may present beneficial effects to plants grown on Earth which experience stress.
Task Progress:	This project was developed to investigate the microbial communities of Veggie grown flight plant samples returned from the ISS. At least one series of plant tissues from each returned plant was sequenced (Veggie 01 and 03) and analyzed including taxonomic identification and inferred function. Bacterial or fungal isolates were further identified to the species level. All sequencing tasks were completed, and all data analyzed. A manuscript is being completed but is currently unpublished.
Bibliography Type:	Description: (Last Updated: 10/29/2025)
Articles in Peer-reviewed Journals	Karouia F, de Oliveira LC, Hameed A, Simpson A, Singh N, Rekha PD, Mason CE, Venkateswaran K. "Genomic characterization of Microbacterium meiriae sp. nov., a novel bacterium isolated from the International Space Station." Sci Rep. 2025 Oct 14;15:35770. https://doi.org/10.1038/s41598-025-13974-5 ; PMID: 41087377; PMCID: PMC12521503 , Oct-2025
Conference Materials (Downloadable)	Khodadad CL, Simpson A, Singh N, Massa GD, Venkatswaran K. "Microbe-Plant interaction with International Space Station microbial communities on Veggie flight crops." American Society for Gravitational and Space Research Annual Conference, Washington DC, November 12-18, 2023. , Nov-2023